

THERMODYNAMICS · TUTORIAL 2

Thermodynamics Tutorial Week 2 — Worked Solutions

Fully worked, independently checked solutions with the original tutorial questions embedded beside the workings. Mathematics is pre-rendered as SVG, so symbols remain readable without MathJax, LaTeX, or an internet connection.

Worked solutions

Original questions embedded

Offline mathematics

PDF-ready

Scope and assumptions These solutions use the values printed on the locally stored 2026 Tutorial 2 slides. SI units are used throughout. Question 4 contains a likely source-unit error in Young's modulus; both the literal printed result and the physically plausible steel result are shown explicitly. Numerical answers were independently recomputed.

Question 1 — Specific kinetic energy

Question 1

1. Determine the specific kinetic energy of a mass whose velocity is 45 m/s, in kJ/kg

Original Thermodynamics Tutorial 2, Question 1

Specific kinetic energy is kinetic energy per unit mass:

$$e_k = \frac{V^2}{2}.$$

With $V = 45 \text{ m/s}$,

$$e_k = \frac{(45)^2}{2} = 1012.5 \text{ J/kg}.$$

Convert to kJ/kg:

$$1012.5 \text{ J/kg} = 1.0125 \text{ kJ/kg}.$$

Therefore,

$$\boxed{e_k = 1.0125 \text{ kJ/kg}}.$$

Question 2 — Energy versus acceleration time

Question 2:

A car is accelerated from rest to 85 km/h in 5 s. Would the energy transferred to the car be different if it were accelerated to the same speed in 3.5 s?

Original Thermodynamics Tutorial 2, Question 2

The car starts from rest and reaches the same final speed in both cases. Neglecting changes in elevation, drag, rolling resistance, and losses, the energy transferred to the car equals its change in kinetic energy:

$$W = \Delta KE = \frac{1}{2}m(V_2^2 - V_1^2).$$

Here $V_1 = 0$ and the same $V_2 = 85 \text{ km/h}$ is reached in both cases. Therefore,

$$W_{5.0 \text{ s}} = W_{3.5 \text{ s}}.$$

The energy is not different, but the average power is:

$$\dot{W}_{\text{avg}} = \frac{W}{\Delta t}.$$

Thus,

$$\frac{\dot{W}_{3.5}}{\dot{W}_{5.0}} = \frac{5.0}{3.5} = 1.4286.$$

So reaching the speed in 3.5 s requires approximately 42.9% more average power than taking 5.0 s, even though the ideal energy transfer is the same.

Question 3 — Sensible heating and heat-transfer rate

Question 3:

A 5 kg system was taken from 50°C to 150°C. Given the specific heat capacity of the system is $C_p = 6000 \text{ J/kg}$, determine;

- How much energy in the form of heat was added to the system to produce this temperature increase?
- If the system received the heat over a period of 10 seconds, calculate the heat transfer rate in kW.

Original Thermodynamics Tutorial 2, Question 3

Given

$$m = 5 \text{ kg}, \quad c_p = 6000 \text{ J/(kg K)}, \quad \Delta T = 150 - 50 = 100 \text{ K}.$$

(a) Heat added

For constant specific heat,

$$Q = mc_p \Delta T.$$

Therefore,

$$Q = (5)(6000)(100) = 3,000,000 \text{ J} = 3000 \text{ kJ}.$$

Hence,

$$\boxed{Q = 3000 \text{ kJ} = 3.00 \text{ MJ}}.$$

(b) Heat-transfer rate over 10 seconds

$$\dot{Q} = \frac{Q}{\Delta t} = \frac{3000 \text{ kJ}}{10 \text{ s}} = 300 \text{ kJ/s}.$$

Since $1 \text{ kJ/s} = 1 \text{ kW}$,

$$\boxed{\dot{Q} = 300 \text{ kW}}.$$

Question 4 — Work required to stretch a steel rod

Question 4:

A steel rod of 0.8 cm diameter and 21 m length is stretched 3 cm. Young's modulus for this steel is 21 kN/cm². How much work, in kJ, is required to stretch this rod?

Original Thermodynamics Tutorial 2, Question 4

For a linearly elastic rod,

$$\sigma = E\varepsilon, \quad \varepsilon = \frac{\delta}{L}, \quad F = \sigma A = EA \frac{\delta}{L}.$$

Because the force rises linearly from zero to F , the work is

$$W = \frac{1}{2} F \delta = \frac{EA \delta^2}{2L}.$$

The rod area is

$$A = \frac{\pi d^2}{4} = \frac{\pi (0.008)^2}{4} = 5.02655 \times 10^{-5} \text{ m}^2.$$

Also,

$$L = 21 \text{ m}, \quad \delta = 0.03 \text{ m}.$$

Literal result using the value printed on the slide

The slide states

$$E = 21 \text{ kN/cm}^2.$$

Since $1 \text{ kN/cm}^2 = 10^7 \text{ Pa}$,

$$E = 2.1 \times 10^8 \text{ Pa} = 210 \text{ MPa}.$$

Then

$$F = EA \frac{\delta}{L} = (2.1 \times 10^8)(5.02655 \times 10^{-5}) \frac{0.03}{21} = 15.0796 \text{ N}.$$

Therefore,

$$W = \frac{1}{2} (15.0796)(0.03) = 0.226195 \text{ J} = 2.26195 \times 10^{-4} \text{ kJ}.$$

Using the slide literally,

$$W = 2.262 \times 10^{-4} \text{ kJ}.$$

Source-data warning

The printed modulus 210 MPa is approximately one thousand times smaller than a typical steel Young's modulus of 210 GPa. If the intended value was

$$E = 21,000 \text{ kN/cm}^2 = 210 \text{ GPa},$$

then the work would be one thousand times larger:

$$W \approx 0.2262 \text{ kJ}.$$

The lecturer or tutor should confirm which modulus was intended.

Question 5 — Shaft torque from power and speed

Question 5:

Determine the torque applied to the shaft of a car that transmits 250 hp (1 hp $\approx$ 745.7 W) and rotates at a rate of 3200 rpm.

Original Thermodynamics Tutorial 2, Question 5

Power and torque are related by

$$\dot{W} = \tau\omega.$$

Convert the transmitted power:

$$\dot{W} = (250 \text{ hp})(745.7 \text{ W/hp}) = 186,425 \text{ W}.$$

Convert 3200 rpm to angular velocity:

$$\omega = (3200 \text{ rev/min}) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) = 335.103 \text{ rad/s}.$$

Thus,

$$\tau = \frac{\dot{W}}{\omega} = \frac{186,425}{335.103} = 556.321 \text{ N m}.$$

Therefore,

$$\boxed{\tau \approx 556.3 \text{ N m}}.$$

Question 6 — Minimum lifting time

Question 6:

A 100 kW engine and an 80 MW power plant are each used to lift a 500,000 kg mass 100 meters vertically. Calculate the minimum time (in seconds) it would take for each power source to complete this task.

Original Thermodynamics Tutorial 2, Question 6

The ideal work required to lift the mass is its increase in gravitational potential energy:

$$W = mgh.$$

With $m = 500,000 \text{ kg}$, $g = 9.81 \text{ m/s}^2$, and $h = 100 \text{ m}$,

$$W = (500,000)(9.81)(100) = 4.905 \times 10^8 \text{ J}.$$

The minimum ideal time is

$$t = \frac{W}{\dot{W}}.$$

100 kW engine

$$t_{100 \text{ kW}} = \frac{4.905 \times 10^8}{100 \times 10^3} = 4905 \text{ s}.$$

Therefore,

$$t_{100 \text{ kW}} = 4905 \text{ s} = 81.75 \text{ min}.$$

80 MW power plant

$$t_{80 \text{ MW}} = \frac{4.905 \times 10^8}{80 \times 10^6} = 6.13125 \text{ s}.$$

Thus,

$$t_{80 \text{ MW}} \approx 6.13 \text{ s}.$$

These are ideal minimum times; real systems would take longer because of efficiency and power-delivery limits.

Question 7 — Closed-system energy balance

Question 7:

A vertical piston-cylinder device contains water and is being heated. During the process, 70 kJ of heat is transferred to the water, and heat losses from the side walls amount to 9 kJ. The piston rises as a result of evaporation, and 6 kJ of work is done by the vapor. Determine the change in the energy of the water for this process.

Original Thermodynamics Tutorial 2, Question 7

Use the closed-system energy balance with heat into the system positive and work done by the system positive:

$$\Delta E = Q - W.$$

The net heat transferred to the water is

$$Q = Q_{\text{in}} - Q_{\text{loss}} = 70 - 9 = 61 \text{ kJ}.$$

The piston does 6 kJ of work on the surroundings, so

$$W = 6 \text{ kJ}.$$

Therefore,

$$\Delta E = 61 - 6 = 55 \text{ kJ}.$$

Hence,

$$\boxed{\Delta E_{\text{water}} = +55 \text{ kJ}}.$$

The positive sign means the water's total energy increased.

Question 8 — Rate of increase of room energy

Question 8

Consider a room that is initially at the outdoor temperature of 30°C. The room contains a 50-W lightbulb, a 120-W TV set, a 300-W refrigerator, and a 1200-W iron. Assuming no heat transfer through the walls, determine the rate of increase of the energy content of the room when all of these electric devices are on.

Original Thermodynamics Tutorial 2, Question 8

With no heat transfer through the walls, all electrical power consumed by the devices ultimately increases the room's energy. Sum the device powers:

$$\frac{dE_{\text{room}}}{dt} = 50 + 120 + 300 + 1200 = 1670 \text{ W}.$$

Therefore,

$$\boxed{\frac{dE_{\text{room}}}{dt} = 1670 \text{ W} = 1.67 \text{ kW}}.$$

The initial room temperature is not needed for the requested energy-rate calculation.

Question 9 — Additional power for vehicle acceleration

Question 9:

Consider a 2500-kg car cruising at constant speed of 80 km/h. Now the car starts to pass another car by accelerating to 110 km/h in 5 s. (a) Determine the additional power needed to achieve this acceleration. (b) What would your answer be if the total mass of the car were only 800 kg?

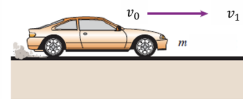


Analysis:

The work needed to accelerate a body is simply the change in the kinetic energy of the body.

$$W_a = \Delta E_k = \frac{1}{2} m(\Delta v^2) = \frac{1}{2} m(v_1^2 - v_0^2)$$

$$\text{and } \dot{W}_a = \frac{W_a}{\Delta t}$$



At t_0 , $v_0 = 80$ km/h

At t_1 , $v_1 = 110$ km/h

Original Thermodynamics Tutorial 2, Question 9

Neglecting losses, the additional work is the change in kinetic energy:

$$W_a = \frac{1}{2} m(V_1^2 - V_0^2).$$

Convert the speeds:

$$V_0 = \frac{80}{3.6} = 22.2222 \text{ m/s},$$

$$V_1 = \frac{110}{3.6} = 30.5556 \text{ m/s}.$$

The average additional power over $\Delta t = 5$ s is

$$\dot{W}_a = \frac{m(V_1^2 - V_0^2)}{2\Delta t}.$$

(a) $m = 2500$ kg

$$\dot{W}_a = \frac{(2500) [(30.5556)^2 - (22.2222)^2]}{2(5)} = 109,953.7 \text{ W}.$$

Therefore,

$$\boxed{\dot{W}_a \approx 109.95 \text{ kW}}.$$

(b) $m = 800$ kg

$$\dot{W}_a = \frac{(800) [(30.5556)^2 - (22.2222)^2]}{2(5)} = 35,185.2 \text{ W}.$$

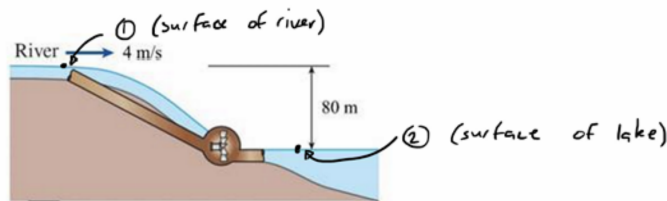
Thus,

$$\boxed{\dot{W}_a \approx 35.19 \text{ kW}}.$$

Question 10 — River mechanical energy and generation potential

Question 10:

Consider a river flowing toward a lake at an average velocity of 4 m/s at a rate of 600 m³/s at a location 80 m above the lake surface. Determine the total mechanical energy of the river water per unit mass and the power generation potential of the entire river at that location.



Original Thermodynamics Tutorial 2, Question 10

For an incompressible stream, total mechanical energy per unit mass is

$$e_{\text{mech}} = \frac{P}{\rho} + \frac{V^2}{2} + gz.$$

Take the lake surface as $z = 0$. Both the river and lake surfaces are exposed to atmospheric pressure, so their gauge pressure is zero. At the river location,

$$V = 4 \text{ m/s}, \quad z = 80 \text{ m}.$$

Therefore,

$$e_{\text{mech}} = 0 + \frac{4^2}{2} + (9.81)(80) = 8 + 784.8 = 792.8 \text{ J/kg}.$$

Thus,

$$e_{\text{mech}} = 0.7928 \text{ kJ/kg}.$$

Using $\rho \approx 1000 \text{ kg/m}^3$ and $\dot{V} = 600 \text{ m}^3/\text{s}$, the mass-flow rate is

$$\dot{m} = \rho \dot{V} = (1000)(600) = 600,000 \text{ kg/s}.$$

The theoretical power-generation potential is

$$\dot{W}_{\text{max}} = \dot{m} e_{\text{mech}}.$$

Hence,

$$\dot{W}_{\text{max}} = (600,000)(792.8) = 4.7568 \times 10^8 \text{ W}.$$

Therefore,

$$\dot{W}_{\text{max}} = 475.68 \text{ MW}.$$

This is an ideal upper bound; real electrical output is lower because of losses.

Final answers at a glance

- **1:** 1.0125 kJ/kg.
- **2:** Same ideal energy; the 3.5 s acceleration requires 1.4286 times the average power.
- **3:** $Q = 3000 \text{ kJ}$, $\dot{Q} = 300 \text{ kW}$.
- **4:** Literal printed modulus: $2.262 \times 10^{-4} \text{ kJ}$; if $E = 210 \text{ GPa}$ was intended: 0.2262 kJ.
- **5:** 556.3 N m.
- **6:** 4905 s at 100 kW; 6.13 s at 80 MW.
- **7:** $\Delta E = +55 \text{ kJ}$.
- **8:** 1.67 kW.
- **9:** 109.95 kW for 2500 kg; 35.19 kW for 800 kg.
- **10:** 0.7928 kJ/kg and 475.68 MW ideal potential.